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FixTrack: Multi-Branch Maintenance and Service Request System

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1. System Overview

1.1. Problem Statement

Organizations with multiple branches often report broken equipment and facility problems through paper forms, phone calls, or separate group chats. These reports can be overlooked, repeated, or passed between staff without a clear person responsible for the repair. Administrators have difficulty identifying urgent issues, checking technician workloads, and determining which requests are still unresolved.

Maintenance records and repair costs are also scattered across branches. Without a shared history, staff may repeatedly repair the same asset without knowing its previous problems or total cost. This makes it difficult to monitor service quality, explain delays, and decide when equipment should be replaced.

1.2. System Description

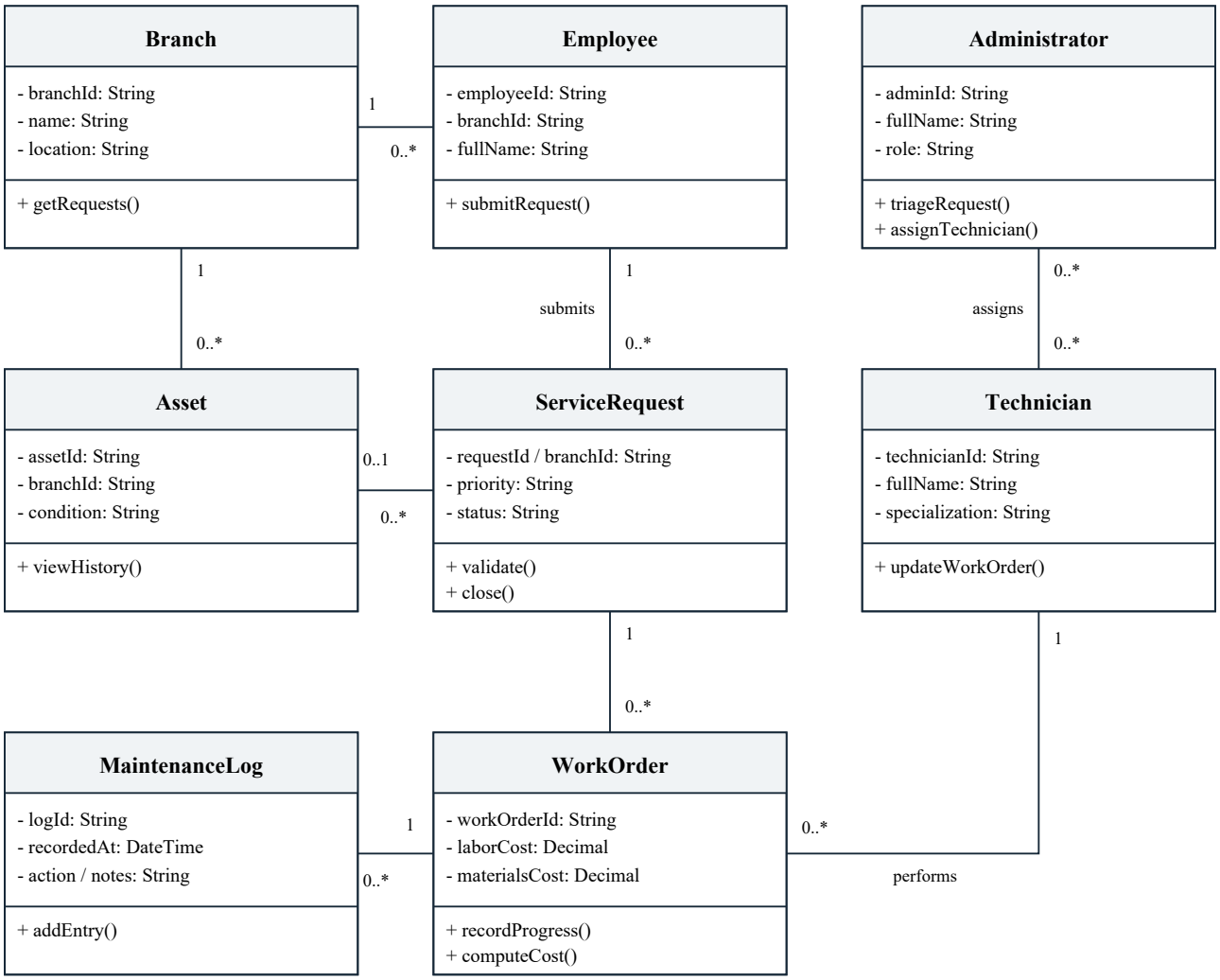
FixTrack is a web-based maintenance and service request system for organizations with multiple branches. Employees submit equipment or facility issues with a description, branch, location, and an optional photo. A request can reference a registered asset, but general facility concerns can be reported without one. Administrators review the report, set its priority, and create a work order assigned to a technician.

Technicians update their assigned work orders, record repair actions, and enter labor and material costs. The administrator verifies the resolution before closing the request. Request and work-order histories retain who made each update and when, allowing staff to trace the full repair process.

A centralized dashboard provides branch-level views of open requests, ongoing repairs, completed work, and recorded costs. Employees see their own requests, technicians see their assigned jobs, and administrators manage the branches within their access. The initial prototype requires an internet connection; preventive scheduling, offline operation, and automated notifications are future extensions.

2. UML Diagrams

2.1. Class Diagram



Association ends show multiplicities. Each work order has one assigned technician.

Figure 2.1. Class Diagram - FixTrack

The class diagram shows the main records and their relationships. A Branch has employees and assets, while an Employee can submit many ServiceRequests. Each request belongs to a branch and may refer to one Asset; a facility issue does not require an asset. An Administrator reviews requests and assigns technicians. A ServiceRequest can produce multiple WorkOrders, allowing follow-up repairs, and each assigned work order is handled by one Technician. WorkOrders contain labor and material costs and have MaintenanceLogs that preserve repair actions and timestamps. Together, these records provide an asset history and support branch reports without deleting earlier updates.

2.2. Activity Diagram

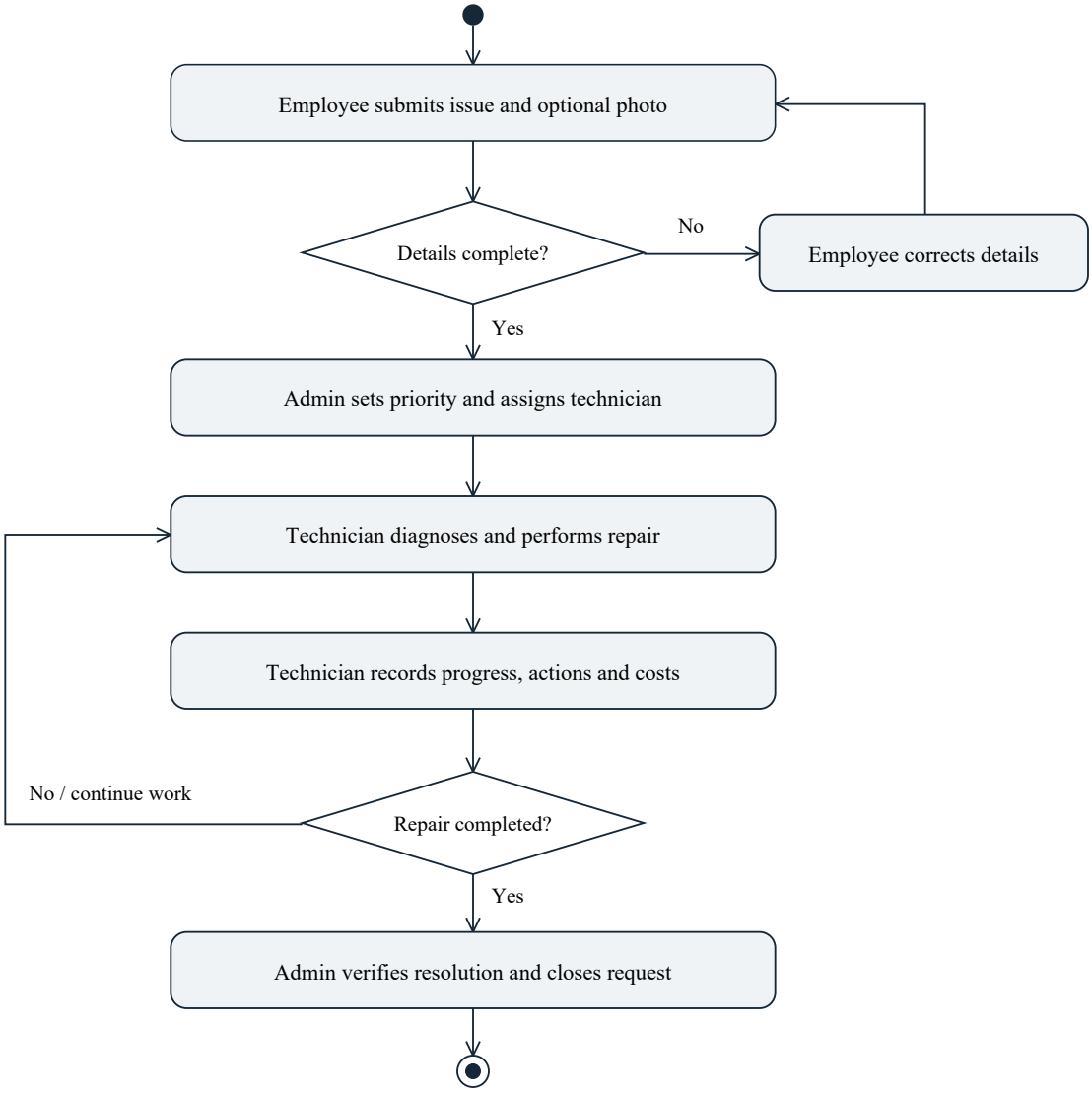
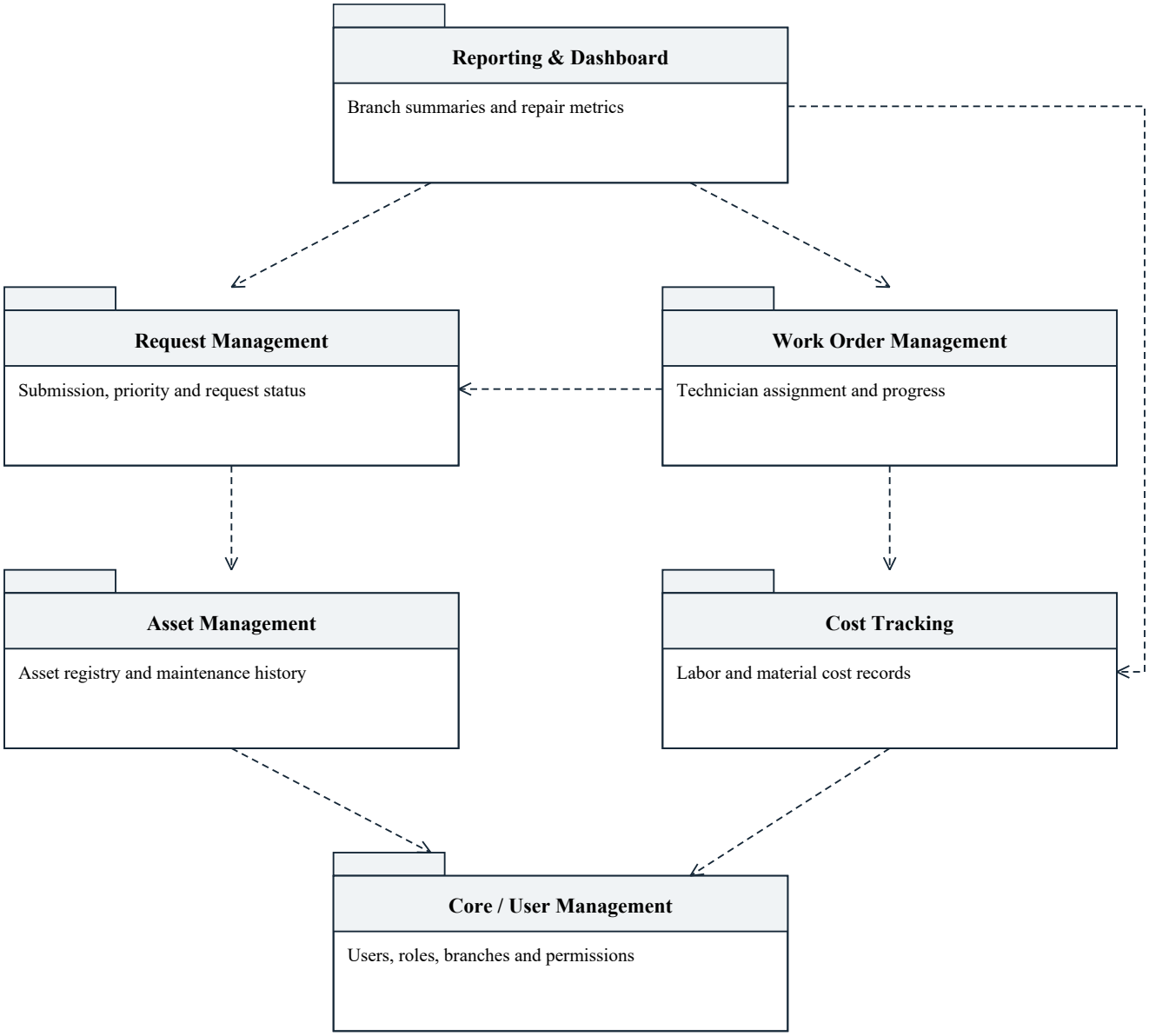


Figure 2.2. Activity Diagram - Service Request and Repair Workflow

The activity diagram follows a request from reporting to closure. The employee submits the issue and corrects any missing information before the report proceeds. The administrator then sets its priority and assigns a technician through a work order. The technician diagnoses the problem, carries out the repair, and records progress, actions, and costs. If the repair is incomplete, work continues and more updates are added. Once the technician reports completion, the administrator verifies the result and closes the request. Status changes remain in the history so users can see what happened and who handled each stage.

2.3. Package Diagram



Dashed arrow: dependent package -> package it uses.

Figure 2.3. Package Diagram - FixTrack Modules and Dependencies

The system is divided into six packages. Request Management handles reports and status, while Work Order Management handles assignments and repair progress. Asset Management stores equipment details and repair history. Cost Tracking records labor and materials against work orders. Reporting & Dashboard reads request, work-order, and cost records to summarize branch activity. Core/User Management supplies shared users, roles, branches, and access rules to all packages; the arrows show key dependencies rather than every shared call. This separation allows team members to develop different modules while keeping their responsibilities clear.

2.4. Deployment Diagram

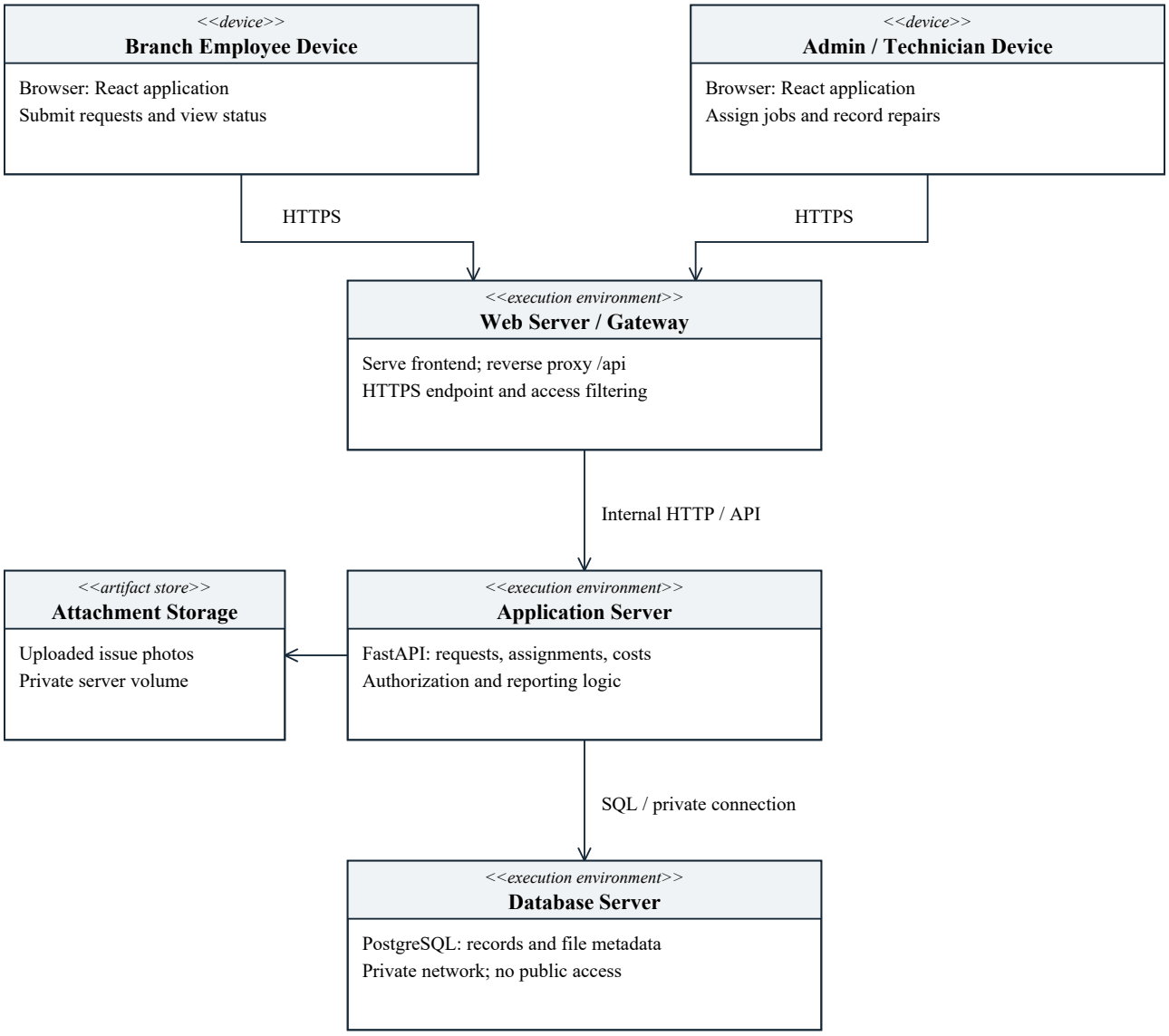
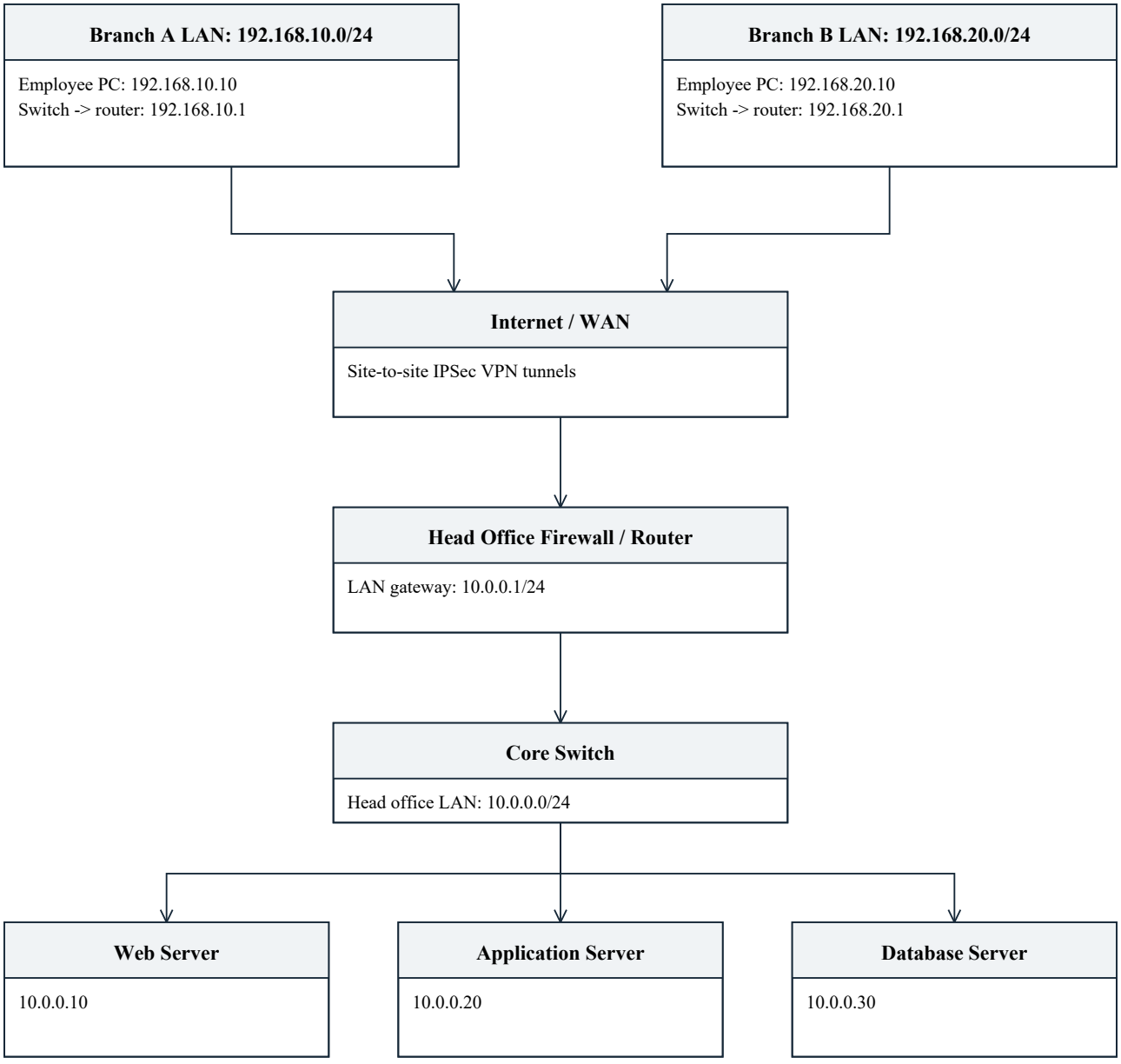


Figure 2.4. Deployment Diagram - FixTrack Runtime Components

Employees, administrators, and technicians access FixTrack through a browser on a computer or mobile device. The web server serves the React interface and forwards API requests to the FastAPI application server, where request validation, assignments, access rules, and reporting run. PostgreSQL stores the structured records, while uploaded photos are kept in private attachment storage with their metadata in the database. This is a proposed production deployment; the classroom prototype can run these logical components on one development computer. No biometric scanners or branch edge servers are required.

2.5. Network Diagram



Private addressing is illustrative; each branch uses a distinct subnet.

Figure 2.5. Network Diagram - Illustrative Multi-Branch Infrastructure

The network diagram illustrates a production option with separate branch subnets connected to head office through IPSec VPN tunnels. Traffic passes through the head-office firewall and core switch to the web, application, and database servers. Browser access uses HTTPS; the database is reachable only by the application service. The addresses are examples, and each branch uses a distinct subnet to avoid routing conflicts. VPN equipment is not a requirement for the classroom prototype: a local development network can demonstrate the same application workflows.

3. Future Expansion

FixTrack could later support preventive maintenance schedules that create work orders before equipment fails. QR codes on assets could let employees open the correct equipment record when reporting an issue. Email or SMS notifications could alert users to assignments, status changes, and overdue repairs, while service targets could help administrators identify requests that need escalation.

Other possible extensions include spare-parts inventory, purchase requests for repair materials, external service-provider coordination, and mobile offline capture with synchronization when connectivity returns. Long-term reports could compare repair costs and recurring failures to support equipment replacement decisions. These are future features rather than requirements of the initial prototype.

4. Project Summary

FixTrack addresses the difficulty of coordinating maintenance across multiple branches. It replaces scattered reports with a single workflow for submitting issues, assigning technicians, recording repair progress and costs, and confirming resolution. Employees can follow their requests, technicians can manage assigned work, and administrators can monitor branch performance through a shared dashboard.

The proposed system has clear user roles, related data entities, and separate modules for requests, work orders, assets, costs, reporting, and user management. Its first version can be developed as a web application with a centralized database, making the project practical for a student team while leaving room for preventive maintenance and other extensions.